

K-01-CX-CKN-smallness-propagation-kill-test v1: CKN smallness vs F -concentration lower bound

ZFL-CX

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1 Mission

$$\boxed{C_F^{conc} + C_{CKN} \xrightarrow{?} \int_0^{T^*} W_+ < \infty}$$

$$\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_K \Rightarrow \perp}$$

where $W_+ = \int (\omega^T S \omega)_+ dx$, $C_F^{conc} : \|u(t_n)\|_{L^3(B_{r_n}(x_n))} \geq \varepsilon_0$, $r_n \downarrow 0$, $t_n \uparrow T^*$.

Goal is not to assume $W_+ \lesssim r_n^{-3}$ by scaling. Goal is to derive

$$\boxed{W_+(t) \leq G(r(t), \varepsilon_0)}, \quad \boxed{\int_0^{T^*} G(r(t), \varepsilon_0) dt < \infty}$$

2 CKN barrier

Standard ε -regularity: there exists $\varepsilon_{CKN} > 0$ universal such that if for some cylinder $Q_r(x_0, t_0) = B_r(x_0) \times (t_0 - r^2, t_0)$,

$$r^{-2} \int_{Q_r} |u|^3 + |p|^{3/2} dx dt \leq \varepsilon_{CKN}$$

then u regular in $Q_{r/2}$.

Key: hypothesis is *smallness* of scale-invariant functional. F-01 provides opposite:

$$\|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0$$

i.e., lower bound at time slice, not space-time smallness. No upper bound. Pressure term p not controlled by F .

Thus cannot apply CKN directly to deduce regularity at (x_n, t_n) . The only information is that outside concentration balls, we might have smallness, but that gives regularity *away* from concentration, not control of W_+ which lives near concentration.

3 Attempted propagation

Assume we try to get smallness in annulus $B_{2r_n} \setminus B_{r_n}$. If $\|u(t_n)\|_{L^3(B_{2r_n})} \leq 2\varepsilon_0$ and lower bound holds in B_{r_n} , then outer annulus might have small L^3 . But ε_0 from C_F^{conc} is lower threshold for concentration, not necessarily $\ll \varepsilon_{CKN}$. In fact ε_0 could be $\geq \varepsilon_{CKN}$, so even smallness fails.

Even if we had $r^{-2} \int_{Q_r} |u|^3 \leq \varepsilon_{CKN}$ for $r \sim r_n$, CKN gives

$$\|u\|_{L^\infty(Q_{r/2})} \leq Cr^{-1}$$

Then naive bound:

$$W_+(t) \lesssim \int_{B_{r/2}} |\nabla u|^3 \lesssim r^{-3} |B_{r/2}| \sim r^0? \text{ Actually } |\nabla u| \lesssim r^{-2} \Rightarrow |\omega|^2 \alpha \lesssim r^{-6}$$

More precise: if $|u| \lesssim r^{-1}$, then $|\omega| \lesssim r^{-2}$, $W_+ \lesssim r^{-3}$ after integration over ball $r^3 \cdot r^{-6} = r^{-3}$.

Thus $W_+(t) \lesssim r_n(t)^{-3}$. Since $r_n \downarrow 0$ as $t \uparrow T^*$, this bound blows up, does *not* give integrability:

$$\int^{T^*} r_n(t)^{-3} dt = +\infty \text{ unless } r_n(t) \gtrsim (T^* - t)^\beta, \beta < 1/3$$

No relation between $r_n(t)$ and $T^* - t$ is provided by C_F^{conc} or C_I^{freq} ; only existence of sequence $r_n \downarrow 0$, no rate.

To get $\int G < \infty$, need $G(r, \varepsilon_0) = o((T^* - t)^{-1})$. But r_n^{-3} gives opposite.

Hence even if CKN gave local L^∞ bound, it worsens $\int W_+$.

4 Kill Test

K-01 KILLED if:

1. F-01 only provides lower bound, CKN needs upper/smallness: we have no $r^{-2} \int |u|^3 \leq \varepsilon_{CKN}$.
2. CKN only gives local regularity away from concentration, not W_+ control near concentration.
3. Local regularity does not yield temporally integrable W_+ estimate: r_n^{-3} not integrable as $r_n \rightarrow 0$.
4. r_n^{-3} appears only by dimensional analysis, without derivation from CKN + C_F^{conc} .
5. Estimate still allows $\int_0^{T^*} W_+ = +\infty$: compatible with blow-up, no $\perp$.

All 1-5 hold. The attempted $G(r, \varepsilon_0) = Cr^{-3}$ fails integrability. No quantitative link $r(t) \sim (T^* - t)^{1/2}$ can be derived from L^3 concentration alone; would require Type I assumption, which is not consequence of $T^* < \infty$ (blocked).

Therefore $C_F^{conc} + C_{CKN}$ does not imply $\int W_+ < \infty$.

5 Conclusion

$$C_K : \exists r(t) \text{ with } r^{-2} \int_{Q_r} |u|^3 \leq \varepsilon_{CKN} \text{ not derivable from } T^* < \infty$$

$$W_+(t) \leq G(r(t), \varepsilon_0) \text{ with } G \sim r^{-3} \text{ not integrable}$$

$$C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_K \not\Rightarrow \perp$$

Classification: CKN is genuinely distinct family, but K-01 KILLED as bridge. Smallness cannot be extracted from lower bound; even if extracted away from concentration, it does not control W_+ near concentration in integrable way.

C_{PROVEN} remains $\{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}$.

No K-02. Line closed.